Core Hours

Knowledge Unit	CS Core	KA Core
Understanding the User	2	5
Accountability and Responsibility in Design	2	2
Accessibility and Inclusive Design	2	2
Evaluating the Design	1	2
System Design	1	5
Society, Ethics, and the Profession	Included in SEP hours	
Total Hours	**8**	**16**

Knowledge Units

HCI-User: Understanding the User: Individual goals and interactions with others

CS Core:

1. User-centered design and evaluation methods. (See also: SEP-Context, SEP-Ethical-Analysis, SEP-Professional-Ethics)
    a. "You are not the users"
    b. User needs-finding
    c. Formative studies
    d. Interviews
    e. Surveys
    f. Usability tests

KA Core:

2. User-centered design methodology. (See also: SE-Tools)
    a. Personas/persona spectrum
    b. User stories/storytelling and techniques for gathering stories
    c. Empathy maps
    d. Needs assessment (techniques for uncovering needs and gathering requirements - e.g., interviews, surveys, ethnographic and contextual enquiry) (See also: SE-Requirements)
    e. Journey maps
    f. Evaluating the design (See also: HCI-Evaluation)
    g. Interfacing with stakeholders, as a team
    h. Risks associated with physical, distributed, hybrid and virtual teams
3. Physical and cognitive characteristics of the user
    a. Physical capabilities that inform interaction design (e.g., color perception, ergonomics)

   b. Cognitive models that inform interaction design (e.g., attention, perception and recognition, movement, memory)
   c. Topics in social/behavioral psychology (e.g., cognitive biases, change blindness)
4. Designing for diverse user populations. (See also: SEP-DEIA, HCI-Accessibility)
   a. How differences (e.g., in race, ability, age, gender, culture, experience, and education) impact user experiences and needs
   b. Internationalization
   c. Designing for users from other cultures
   d. Cross-cultural design
   e. Challenges to effective design evaluation. (e.g., sampling, generalization; disability and disabled experiences)
   f. Universal design
5. Collaboration and communication (See also: AI-SEP, SE-Teamwork, SEP-Communication, SPD-Game)
   a. Understanding the user in a multi-user context
   b. Synchronous group communication (e.g., chat rooms, conferencing, online games)
   c. Asynchronous group communication (e.g., email, forums, social networks)
   d. Social media, social computing, and social network analysis
   e. Online collaboration
   f. Social coordination and online communities
   g. Avatars, characters, and virtual worlds

Non-core:

6. Multi-user systems

Illustrative Learning Outcomes:

CS Core:

1. Conduct a user-centered design process that is integrated into a project.

KA Core:

2. Compare and contrast the needs of users with those of designers.
3. Identify the representative users of a design and discuss who else could be impacted by it.
4. Describe empathy and evaluation as elements of the design process.
5. Carry out and document an analysis of users and their needs.
6. Construct a user story from a needs assessment.
7. Redesign an existing solution to a population whose needs differ from those of the initial target population.
8. Contrast the different needs-finding methods for a given design problem.
9. Reflect on whether your design would benefit from low-tech or no-tech components.

Non-core:

10. Recognize the implications of designing for a multi-user system/context.

HCI-Accountability: Accountability and Responsibility in Design

CS Core: (See also: SEP-Context)

1. Design impact
    a. Sustainability (See also: SEP-Sustainability)
    b. Inclusivity (See also: SEP-DEIA)
    c. Safety, security and privacy (See also: SEP-Security, SEC-Foundations)
    d. Harm and disparate impact (See also: SEP-DEIA)
2. Ethics in design methods and solutions (See also: SEP-Ethical-Analysis, SEP-Context, SEP-Intellectual Property)
    a. The role of artificial intelligence (See also: AI-SEP)
    b. Responsibilities for considering stakeholder impact and human factors (See also: SEP-Professional-Ethics)
    c. Role of design to meet user needs
3. Requirements in design (See also: SEP-Professional-Ethics)
    a. Ownership responsibility
    b. Legal frameworks, compliance requirements
    c. Consideration beyond immediate user needs, including via iterative reconstruction of problem analysis and "digital well-being" features

KA Core:

4. Value-sensitive design (See also: SEP-Ethical-Analysis, SEP-Context, SEP-Communication)
    a. Identify direct and indirect stakeholders
    b. Determine and include diverse stakeholder values and value systems.
5. Persuasion through design (See also: SEP-Communication)
    a. Assess the persuasive content of a design
    b. Employ persuasion as a design goal
    c. Distinguish persuasive interfaces from manipulative interfaces

Illustrative Learning Outcomes:

CS Core:

1. Identify and critique the potential impacts of a design on society and relevant communities to address such concerns as sustainability, inclusivity, safety, security, privacy, harm, and disparate impact.

KA Core:

2. Identify the potential human factor elements in a design.
3. Identify and understand direct and indirect stakeholders.
4. Develop scenarios that consider the entire lifespan of a design, beyond the immediately planned uses that anticipate direct and indirect stakeholders.
5. Identify and critique the potential factors in a design that impact direct and indirect stakeholders and broader society (e.g., transparency, sustainability of the system, trust, artificial intelligence).
6. Assess the persuasive content of a design and its intent relative to user interests.
7. Critique the outcomes of a design given its intent.
8. Understand the impact of design decisions.

HCI-Accessibility: Accessibility and Inclusive Design

CS Core:

1. Background (See also: SEP-DEIA, SEP-Security)
    a. Societal and legal support for and obligations to people with disabilities
    b. Accessible design benefits everyone
2. Techniques
    a. Accessibility standards (e.g., Web Content Accessibility Guidelines) (See also: SPD-Web)
3. Technologies (See also: SE-Tools)
    a. Features and products that enable accessibility and support inclusive development by designers and engineers
4. IDFs (Inclusive Design Frameworks) (See also: SEP-DEIA)
    a. Recognizing differences
5. Universal design

KA Core:

6. Background
    a. Demographics and populations (permanent, temporary, and situational disability)
    b. International perspectives on disability (See also: SEP-DEIA)
    c. Attitudes towards people with disabilities (See also: SEP-DEIA)
7. Techniques
    a. UX (user experience) design and research
    b. Software engineering practices that enable inclusion and accessibility. (See also: SEP-DEIA)
8. Technologies
    a. Examples of accessibility-enabling features, such as conformance to screen readers
9. Inclusive Design Frameworks
    a. Creating inclusive processes such as participatory design
    b. Designing for larger impact

Non-core:

10. Background (See also: SEP-DEIA)
    a. Unlearning and questioning
    b. Disability studies
11. Technologies: the Return On Investment (ROI) of inclusion
12. Inclusive Design Frameworks: user-sensitive inclusive design (See also: SEP-DEIA)
13. Critical approaches to HCI (e.g., inclusivity) (See also: SEP-DEIA)

Illustrative Learning Outcomes:

CS Core:

1. Identify accessibility challenges faced by people with different disabilities and specify the associated accessible and assistive technologies that address them. (See also: AI-Agents, AI-Robotics)
2. Identify appropriate inclusive design approaches, such as universal design and ability-based design.
3. Identify and demonstrate understanding of software accessibility guidelines.
4. Demonstrate recognition of laws and regulations applicable to accessible design.

KA Core:

5. Apply inclusive frameworks to design, such as universal design and usability and ability-based design, and demonstrate accessible design of visual, voice-based, and touch-based UIs.
6. Demonstrate understanding of laws and regulations applicable to accessible design.
7. Demonstrate understanding of what is appropriate and inappropriate high level of skill during interaction with individuals from diverse populations.
8. Analyze web pages and mobile apps for current standards of accessibility.

Non-core:

9. Biases towards disability, race, and gender have historically, either intentionally or unintentionally, informed technology design.
    a. Find examples.
    b. Consider how those experiences (learnings?) might inform design.
10. Conceptualize user experience research to identify user needs and generate design insights.

HCI-Evaluation: Evaluating the Design

CS Core:

1. Methods for evaluation with users
    a. Formative (e.g., needs-finding, exploratory analysis) and summative assessment (e.g., functionality and usability testing)
    b. Elements to evaluate (e.g., utility, efficiency, learnability, user satisfaction, affective elements such as pleasure and engagement)
    c. Understanding ethical approval requirements before engaging in user research (See also: SE-Tools, SEP-Ethical-Analysis, SEP-Security, SEP-Privacy, SEP-Professional-Ethics)

KA Core:

2. Methods for evaluation with users (See also: SE-Validation)
    a. Qualitative methods (qualitative coding and thematic analysis)
    b. Quantitative methods (statistical tests)
    c. Mixed methods (e.g., observation, think-aloud, interview, survey, experiment)
    d. Presentation requirements (e.g., reports, personas)
    e. User-centered testing
    f. Heuristic evaluation
    g. Challenges and shortcomings to effective evaluation (e.g., sampling, generalization)
3. Study planning
    a. How to set study goals
    b. Hypothesis design
    c. Approvals from Institutional Research Boards and ethics committees (See also: SEP-Ethical-Analysis, SEP-Security, SEP-Privacy)
    d. How to pre-register a study
    e. Within-subjects vs between-subjects design
4. Implications and impacts of design with respect to the environment, material, society, security, privacy, ethics, and broader impacts. (See also: SEC-Foundations)

a. The environment
b. Material
c. Society
d. Security
e. Privacy
f. Ethics
g. Broader impacts

Non-core:

5. Techniques and tools for quantitative analysis
    a. Statistical packages
    b. Visualization tools
    c. Statistical tests (e.g., ANOVA, t-tests, post-hoc analysis, parametric vs non-parametric tests)
    d. Data exploration and visual analytics; how to calculate effect size.
6. Data management
    a. Data storage and data sharing (open science)
    b. Sensitivity and identifiability.

Illustrative Learning Outcomes:

CS Core:

1. Discuss the differences between formative and summative assessment and their role in evaluating design

KA Core:

2. Select appropriate formative or summative evaluation methods at different points throughout the development of a design.
3. Discuss the benefits of using both qualitative and quantitative methods for evaluation.
4. Evaluate the implications and broader impacts of a given design.
5. Plan a usability evaluation for a given user interface, and justify its study goals, hypothesis design, and study design.
6. Conduct a usability evaluation of a given user interface and draw defensible conclusions given the study design.

Non-core:

7. Select and run appropriate statistical tests on provided study data to test for significance in the results.
8. Pre-register a study design, with planned statistical tests.

HCI-Design: System Design

CS Core:

1. Prototyping techniques and tools
    a. Low-fidelity prototyping
    b. Rapid prototyping
    c. Throw-away prototyping

   d. Granularity of prototyping
2. Design patterns
   a. Iterative design
   b. Universal design (See also: SEP-DEIA)
   c. Interaction design (e.g., data-driven design, event-driven design)
3. Design constraints
   a. Platforms (See also: SPD-Game)
   b. Devices
   c. Resources
   d. Balance among usability, security and privacy (See also: SEC-Foundations)

KA Core:

4. Design patterns and guidelines
   a. Software architecture patterns
   b. Cross-platform design
   c. Synchronization considerations
5. Design processes (See also: SEP-Communication)
   a. Participatory design
   b. Co-design
   c. Double-diamond
   d. Convergence and divergence
6. Interaction techniques (See also: GIT-Interaction)
   a. Input and output vectors (e.g., gesture, pose, touch, voice, force)
   b. Graphical user interfaces
   c. Controllers
   d. Haptics
   e. Hardware design
   f. Error handling
7. Visual UI design (See also: GIT-Visualization)
   a. Color
   b. Layout
   c. Gestalt principles

Non-core:

8. Immersive environments (See also: GIT-Immersion)
   a. XR (encompasses virtual reality, augmented reality, and mixed reality)
   b. Spatial audio
9. 3D printing and fabrication
10. Asynchronous interaction models
11. Creativity support tools
12. Voice UI designs

Illustrative Learning Outcomes:

CS Core:

1. Propose system designs tailored to a specified appropriate mode of interaction.
2. Follow an iterative design and development process that incorporates the following:
   a. Understanding the user
   b. Developing an increment
   c. Evaluating the increment
   d. Feeding those results into a subsequent iteration
3. Explain the impact of changing constraints and design tradeoffs (e.g., hardware, user, security.) on system design.

KA Core:

4. Evaluate architectural design approaches in the context of project goals.
5. Identify synchronization challenges as part of the user experience in distributed environments.
6. Evaluate and compare the privacy implications behind different input techniques for a given scenario.
7. Explain the rationale behind a UI design based on visual design principles.

Non-core:

8. Evaluate the privacy implications within a VR/AR/MR scenario

HCI-SEP: Society, Ethics, and the Profession

CS Core:

1. Universal and user-centered design (See also: HCI-User, SEP-DEIA)
2. Accountability (See also: HCI-Accountability)
3. Accessibility and inclusive design (See also: SEP-DEIA, SEP-Security)
4. Evaluating the design (See also: HCI-Evaluation)
5. System design (See also: HCI-Design)

KA Core:

6. Participatory and inclusive design processes
7. Evaluating the design: Implications and impacts of design: with respect to the environment, material, society, security, privacy, ethics, and broader impacts (See also: SEC-Foundations, SEP-Privacy)

Non-core:

8. VR/AR/MR scenarios

Illustrative Learning Outcomes:

CS Core:

1. Conduct a user-centered design process that is integrated into a project.
2. Identify and critique the potential impacts of a design on society and relevant communities to address such concerns as sustainability, inclusivity, safety, security, privacy, harm, and disparate impact.

KA Core:

2. Critique a recent example of a non-inclusive design choice, its societal implications, and propose potential design improvements.
3. Evaluating the design: Identify the implications and broader impacts of a given design.

Non-core:

4. Evaluate the privacy implications within a VR/AR/MR scenario.

Professional Dispositions

- **Adaptable:** An HCI practitioner should be adaptable to address dynamic changes in technology, user needs, and design challenges.
- **Meticulous:** An HCI practitioner should be meticulous in ensuring that their products are both user-friendly and meet the objectives of the project.
- **Empathetic:** An HCI practitioner must demonstrate understanding of the user's needs.
- **Team-oriented:** The successful HCI practitioner should focus on the success of the team.
- **Inventive:** An HCI practitioner should design solutions that are informed by past practice, the needs of the audience, and HCI fundamentals. Creativity is required to blend these into something that solves the problem appropriately and elegantly.

Mathematics Requirements

Required:

- Basic statistics (MSF-Statistics) to support the evaluation and interpretation of results, including central tendency, variability, frequency distribution.

Course Packaging Suggestions

Introduction to HCI for CS majors and minors, to include the following:

- HCI-User: Understanding the User (7 hours)
- HCI-Accountability: Accountability and Responsibility in Design: (2 hours)
- HCI-Accessibility: Accessibility and Inclusive Design: (4 hours)
- HCI-Evaluation: Evaluating the Design: (3 hours)
- HCI-Design: System Design: (10 hours)
- HCI-SEP: Society, Ethics, and the Profession: (2 hours)

Prerequisites:

- CS2

Course objectives: A student who completes this course should be able to describe user-centered design principles and apply them in the context of a small project.

Introduction to Data Visualization to include the following:

- GIT-Visualization (30 hours)
- GIT-Rendering (10 hours)
- HCI-User: Understanding the User (3 hours)